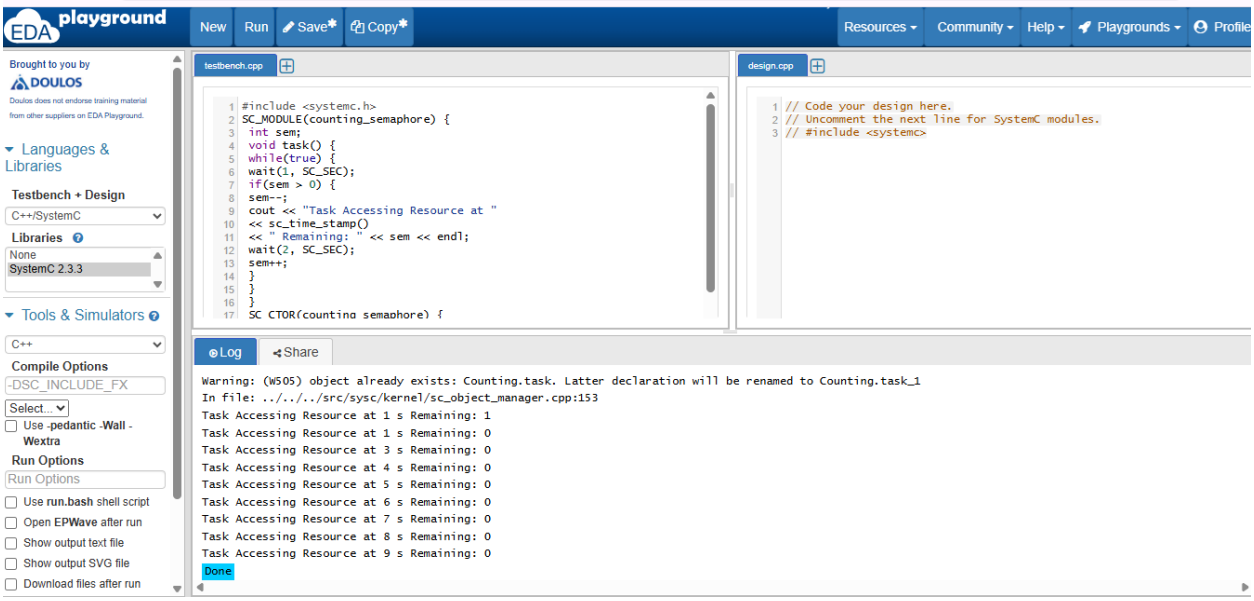
Embedded systems exp 28

```
#include <systemc.h>
SC_MODULE(counting_semaphore) {
 int sem;
 void task() {
 while(true) {
 wait(1, SC_SEC);
 if(sem > 0) {
 sem--;
 cout << "Task Accessing Resource at "
 << sc_time_stamp()
 << " Remaining: " << sem << endl;
 wait(2, SC_SEC);
 sem++;
 }
 }
 }
 SC_CTOR(counting_semaphore) {
 sem = 2;
 SC_THREAD(task);
 SC_THREAD(task);
 SC_THREAD(task);
 }
};
int sc_main(int argc, char* argv[]) {
 counting_semaphore obj("Counting");
 sc_start(10, SC_SEC);
```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testbench.cpp
#include <systemc.h>
SC_MODULE(counting_semaphore) {
int sem;
void task() {
while(true) {
wait(1, SC_SEC);
if(sem > 0) {
sem--;
cout << "Task Accessing Resource at "
<< sc_time_stamp()
<< " Remaining: " << sem << endl;
wait(2, SC_SEC);
sem++;
}
}
}
SC_CTOR(counting_semaphore) {
design.cpp
// Code your design here.
// Uncomment the next line for SystemC modules.
// #include <systemc>
Log
Share
Warning: (W505) object already exists: Counting.task. Latter declaration will be renamed to Counting.task_1
In file: ../../../src/sysc/kernel/sc_object_manager.cpp:153
Task Accessing Resource at 1 s Remaining: 1
Task Accessing Resource at 1 s Remaining: 0
Task Accessing Resource at 3 s Remaining: 0
Task Accessing Resource at 4 s Remaining: 0
Task Accessing Resource at 5 s Remaining: 0
Task Accessing Resource at 6 s Remaining: 0
Task Accessing Resource at 7 s Remaining: 0
Task Accessing Resource at 8 s Remaining: 0
Task Accessing Resource at 9 s Remaining: 0
Done